Mennatallah Abouelenin

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Education

Bilkent University | Bachelor's Degree in Computer Engineering *Ankara, Türkiye | Sep 2021 - Jun 2026*
One of Türkiye's leading research universities; ABET-accredited Computer Engineering program.

Coral International School | High School Diploma *Jeddah, KSA | Sep 2017 - Apr 2021*
Valedictorian, CGPA: 4.0/4.0

Experience

Mobile Application Developer (Intern) *Oasis Technology| Jul 2025 - Sep 2025*
- Built a cross-platform EV charging mobile app using React Native, TypeScript, Expo, and NativeWind, delivering a responsive and modern UI.
- Developed multiple core screens including Home, Map, Route Planning, Booking, Notifications, and Station Details, optimizing usability and app functionality.
- Integrated real-time features such as live location tracking, in-app notifications, and interactive route visualization.

AI Engineer (Intern) *Bluesense AI| Jun 2025 - Jul 2025*
- Implemented deep learning models for skincare analysis using Python, focusing on wrinkle detection, classification, and segmentation.
- Optimized a PyTorch-based CNN and Mask R-CNN pipeline for region-specific wrinkle analysis, improving model accuracy and interpretability.

Front-end Developer *Socavio| Apr 2024 - Feb 2025*
- Utilized Dart and Flutter to design and implement user-friendly UI for a messaging app.
- Developed key application screens including Home, Chat, Add Friend, and Profile pages.
- Managed project tasks and workflow using Jira and maintained code integrity and version control through Git.

Projects

Senior Design Project: 'Sproutly'
- Tech Stack & Tools: Flutter (Dart), FastAPI (Python), SQLite, OpenAI API, Firebase, Swift, Rive, Figma, Git.
- Developed a cross-platform AI-powered productivity application featuring intelligent task management, personalized focus sessions, app blocking, and a gamified virtual companion system.
- Integrated LLM-powered features to automate task breakdown and evaluate user-submitted proof of completed work.
- Led the end-to-end design and implementation of the gamification system and virtual companion, developing achievement tracking, progression mechanics, and companion evolution to encourage long-term user engagement.

Game Jam Project: 'MemoryTrip'
- Tech Stack & Tools: Unity, C#, Git.
- Designed and developed a mobile puzzle game prototype within a 48-hour game jam.
- Implemented core gameplay mechanics, including player movement, interactive puzzle elements, and game logic.

AI Image Classification Project
- Tech Stack & Tools: Python, NumPy, OpenCV, scikit-learn, Matplotlib, Google Colab.
- Developed an image classification system to identify insect species using image preprocessing and feature extraction, and implemented CNN, Random Forest, and SVM classifiers to compare performance and accuracy.

Object-Oriented Software Engineering Project
- Tech Stack & Tools: Python, Django, JavaScript, React, PostgreSQL, CSS, Figma, Git.
- Developed and designed a full-stack web application for Bilkent University's Information Office to manage high school campus tour requests, university fair applications, and general inquiries.
- Implemented CRUD operations, RESTful APIs, and role-based approval workflows connecting the Django backend with the React frontend.

Database Systems Project
- Tech Stack & Tools: Python, Django, TypeScript, Next.js, shadcn/ui, Node.js, PostgreSQL, Docker, CSS, Jira, Figma, Git.
- Built a full-stack second-hand vehicle marketplace system that enables efficient listings, transactions, and data management for vehicle sellers and buyers, and led the UI design and front-end development for all web pages.

Programming Language Project

- Created a programming language for IoT devices, including lexical analysis and core parsing components.

Platform Game Project

- Tech Stack & Tools: Java, LibGDX, GUI, Git.
- Implemented and designed a 2D platform game using object-oriented principles to offer an entertaining and educational experience, helping new students familiarize themselves with the CS department environment.

Skills

Programming Languages: C++, Java, Dart, Python, JavaScript, TypeScript, HTML/CSS, C, C#, SQL, Swift, SystemVerilog, R, Kotlin, PHP
Frameworks & Libraries: Flutter, React, React Native, Django, FastAPI, PyTorch, Unity, Next.js, Node.js, LibGDX
Tools & Technologies: Visual Studio, Android Studio, RStudio, MATLAB, Xcode, Git, Jira, Docker, Vivado, Google Colab, Gradio, Figma, Rive, Adobe Illustrator, Adobe Photoshop, Microsoft Office
Hardware & Digital Systems: Computer Architecture, Digital Design, FPGA Development (Basys 3), IT Support & Troubleshooting
Soft Skills: Problem-solving, Teamwork, Communication, Public speaking, Critical thinking, Adaptability, Creativity

Relevant Coursework

Software Verification & Validation, Engineering Project Management, Machine Learning, Data Science, Bioinformatics, Operating Systems, Graphic Design

Languages

English and Arabic (*Native Proficiency*), French (*Intermediate Proficiency*), Turkish (*Limited Working Proficiency*)

Volunteer Experience

- Completed 50 hours of community service at University of Business & Technology (2021).
- Volunteered in the "Disability Awareness 2020" and the "International Children's Day 2020" events.

Interests

Game development, Animation, Creative Writing, Reading, Tutoring